AC444
VALUATION
AND SECURITES ANALYSIS

Week 7 Session 2
Implications for Price Multiples

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TOPICS DISCUSSED

- Implications of the PVAE for PB Ratios;
- Empirical Evidence on PB Ratios;
- Implications of the PVAE for Trailing PE Ratios;
- Empirical Evidence on Trailing PE Ratios;
- Implications of the PVAE for Leading PE Ratios;
- Strategic Taxonomy and the Joint Distribution of PB and Trailing PE Ratios.

DETERMINANTS OF PRICE-TO-BOOK RATIOS

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DETERMINANTS OF PRICE-TO-BOOK RATIOS

- The PVAE valuation formula can be rewritten to express a firm's (levered) ratio of intrinsic to book value of equity:

$$\frac{VE_t^*}{BVE_t} = 1 + \frac{1}{BVE_t} \sum_{i=1}^{\infty} \frac{E_t(AE_{t+i})}{(1+r_E)^i}$$

with:

- VE_t^* denoting the intrinsic (ex-dividend) value of the firm's equity at date t;
- BVE_t denoting the book value of equity at date t;
- $E_t(AE_{t+i})$ denoting the abnormal earnings expected at date t+i as of date t;
- r_E denoting the cost of equity capital.

DETERMINANTS OF PRICE-TO-BOOK RATIOS

- A firm's (levered) price-to-book ratio thus depends on:
  - ✓ the level of the firm's future abnormal earnings;
  - ✓ the firm's current book value of equity;
  - ✓ the cost of equity capital.

DETERMINANTS OF PRICE-TO-BOOK RATIOS

- Other things being equal:
  - ✓ The higher the level of future expected abnormal earnings, the higher the price-to-book ratio (assuming a non-negative book value of equity);
  - ✓ The lower the book value of equity, the higher the price-to-book ratio (assuming a non-negative present value of expected abnormal earnings);
  - ✓ The lower the cost of equity capital, the higher the price-to-book ratio (assuming a non-negative book value of equity and non-negative present value of expected abnormal earnings).

DETERMINANTS OF PRICE-TO-BOOK RATIOS

- Consider a firm without any source of comparative advantage within a very competitive industry;
- Further assume that the accounting is unbiased;
- Such a firm's intrinsic (levered) price-to-book ratio should thus be equal to 1;
- This price-to-book ratio is referred to as the normal (levered) price-to-book ratio.

DETERMINANTS OF PRICE-TO-BOOK RATIOS

- Consider any expected abnormal earnings generated at date t+i;

- This abnormal earning can be rewritten as:

$$AE_{t+i} = (ROE_{t+i} - r_E)\, BVE_{t+i-1} = AROE_{t+i}\, BVE_{t+i-1}$$

- Furthermore, the book value of equity BVE_{t+i-1} can be rewritten as the current book value of equity BVE_t multiplied by the product of 1 plus the annual growth rates in the book value of equity up to BVE_{t+i-1}.

DETERMINANTS OF PRICE-TO-BOOK RATIOS

- It follows that a firm's (levered) PB ratio thus depends on:
  - ✓ the level of future abnormal profitability, as captured by AROE;
  - ✓ annual growth in the book value of equity;
  - ✓ the cost of equity capital.

DETERMINANTS OF PRICE-TO-BOOK RATIOS

- Other things being equal (assuming non-negative abnormal profitability):
  - the higher the level of future expected abnormal profitability, the higher the price-to-book ratio;
  - the higher the growth value in the book value of equity, the higher the price-to-book ratio;
  - the lower the cost of equity capital, the higher the price-to book ratio.

PRICE-TO-BOOK RATIOS

- The valuation task is thus framed in terms of 2 key questions about a firm's value drivers:
  - ✓ How much greater or smaller than normal will the firm's ROE be?
  - ✓ How quickly will the firm's investment base (book value of equity) grow?

PRICE-TO-BOOK RATIOS

- If desired, the PB expression can be rewritten so that future ROE are expressed as the product of their components:
  - ✓ ROS;
  - ✓ AT;
  - ✓ FL.

APPLICATION EXERCISE 1

- Caelum Inc. trades for 3 years before being liquidated;
- Its cost of equity capital is 10%;

($m)	Year 1	Year 2	Year 3
BVE (BY)	90	60	50
CE	25	35	40

- Assuming efficient markets, calculate Caelum Inc's implied price-to-book ratio at the beginning of year 1.

APPLICATION EXERCISE 1 (SOLUTION)

		1	2	3
BVE(BY)		90	60	50
CE		25	35	40
rE		10%	10%	10%
AEi		16	29	35
PVAEi		14.5	24.0	26.3
PVAE	64.8			
BVE	90.0			
VE	154.8			
PB	**1.72**			

EMPIRICAL EVIDENCE ON PRICE-TO-BOOK RATIOS

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EMPIRICAL EVIDENCE ON LEVERED PRICE-TO-BOOK RATIOS

- Penman ranks all NYSE, AMEX, and NASDAQ stocks on the basis of their PB ratios and allocates them to 20 portfolios;

- He then computes subsequent abnormal earnings (deflated by the book value of equity in the year of allocation to portfolios).

EMPIRICAL EVIDENCE ON LEVERED PRICE-TO-BOOK RATIOS

	PB	0	1	2	3	4	5
P1	6.7	0.18	0.23	0.22	0.22	0.23	0.24
P2	4.0	0.13	0.16	0.14	0.15	0.15	0.14
P5	2.3	0.08	0.08	0.07	0.07	0.08	0.07
P10	1.4	0.03	0.03	0.03	0.04	0.05	0.03
P15	1.0	0.01	0.01	0.01	0.02	0.03	0.02
P19	0.6	–0.05	–0.05	–0.04	–0.01	–0.00	–0.01
P20	0.4	–0.09	–0.07	–0.07	–0.04	–0.02	–0.04

EMPIRICAL EVIDENCE ON LEVERED PRICE-TO-BOOK RATIOS

- On average:
  - ✓ high PB stocks subsequently yield high abnormal earnings;
  - ✓ low PB stocks subsequently yield low abnormal earnings;
  - ✓ Stocks trading on PB ratios close to 1 subsequently experience abnormal earnings which are close to 0.

DETERMINANTS OF TRAILING PRICE-TO-EARNINGS RATIOS

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DETERMINANTS OF TRAILING PRICE-TO-EARNINGS RATIOS

- Starting with the PVAE valuation formula, and editing it, one gets the following formula for intrinsic (levered) cum-dividend trailing price-to earnings ratios:

$$\frac{VE_t^* + ND_t}{CE_t} = \frac{1 + r_E}{r_E}\left[1 + \frac{1}{CE_t}\sum_{i=1}^{\infty}\frac{E_t(AEG_{t+i})}{(1+r_E)^i}\right]$$

with:

- ✓ $VE_t^* + ND_t$ denoting the cum-dividend value of equity at date t;
- ✓ AEG_{t+i} denoting the growth (change) in abnormal earnings from date t-i-1 to date t ($AE_{t+i} - AE_{t+i-1}$).

DETERMINANTS OF TRAILING PRICE-TO-EARNINGS RATIOS

- A firm's cum-dividend (levered) trailing price-to-earnings ratio thus depends on:
  - ✓ expected changes in abnormal earnings;
  - ✓ the level of current comprehensive earnings;
  - ✓ the cost of equity capital.

DETERMINANTS OF TRAILING PRICE-TO-EARNINGS RATIOS

- The higher the expected changes in future abnormal earnings, the higher the PE ratio;

- The higher the current comprehensive earnings, the lower the PE ratio (assuming non-negative expected changes in abnormal earnings);

- The higher the cost of equity capital, the lower the PE ratio (assuming non-negative expected changes in abnormal earnings).

DETERMINANTS OF TRAILING PRICE-TO-EARNINGS RATIOS

- Consider a firm without any source of comparative advantage within a very competitive industry;
- Further assume that accounting is unbiased;
- Such a firm should trade on a trailing (levered) cum-dividend price-to-earnings ratio equal to $(1+r_E)/r_E$ in efficient capital markets;
- This ratio is referred to as the normal (levered) cum-dividend trailing price-to-earnings ratio.

APPLICATION EXERCISE 2

- Consider Caelum Inc. again;
- Calculate Caelum Inc's intrinsic (levered) trailing cum-dividend PE ratio at the end of year 1.

APPLICATION EXERCISE 2

Year		**1**	**2**	**3**	**4**
AEi		16	29	35	0
ΔAEi			13	6	-35
PV(ΔAEi)			11.8	5.0	-26.3
PV(Sum of ΔAEi)/CE1	-0.38				
(1+rE)/rE	11				
Trailing PE	**6.81**				

EMPIRICAL EVIDENCE ON LEVERED TRAILING PRICE-TO-EARNINGS RATIOS

EMPIRICAL EVIDENCE ON LEVERED TRAILING PRICE-TO-EARNINGS RATIOS

- Penman investigates and reports on the relation between PE ratios and growth in abnormal earnings for 3 groups of NYSE and AMEX stocks over a long period of time.

EMPIRICAL EVIDENCE ON LEVERED TRAILING PRICE-TO-EARNINGS RATIOS

Abnormal Earnings

(deflated by the book value of equity in the year of allocation of stocks to portfolios)

		Year				
PE level	**PE**	**0**	**1**	**2**	**3**	**4**
High	25.6	0.9%	3.4%	3.8%	5.4%	6.1%
Medium	10.6	4.0%	4.3%	4.3%	4.4%	5.1%
Low	5.2	5.8%	4.1%	3.5%	3.6%	4.1%

EMPIRICAL EVIDENCE ON LEVERED TRAILING PRICE-TO-EARNINGS RATIOS

- On average:
  - ✓ High PE firms experienced low abnormal earnings (and hence comprehensive earnings?) in the year of allocation of stocks to portfolios and positive growth in abnormal earnings subsequently;
  - ✓ Low PE firms experienced high abnormal earnings (and hence comprehensive earnings?) in the year of allocation of stocks to portfolios and negative growth in abnormal earnings subsequently.

DETERMINANTS OF LEADING PRICE-TO-EARNINGS RATIOS

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DETERMINANTS OF LEADING PRICE-TO-EARNINGS RATIOS

- Starting with the PVAE valuation formula, and editing it, one gets the following formula for intrinsic (levered) leading price-to earnings ratios:

$$\frac{VE_t^*}{CE_{t+1}} = \frac{1}{r_E}\left[1 + \frac{1}{CE_{t+1}}\sum_{i=1}^{\infty}\frac{E_t(AEG_{t+i+1})}{(1+r_E)^i}\right]$$

DETERMINANTS OF LEADING PRICE-TO-EARNINGS RATIOS

- A stock's leading price-to-earnings ratio thus depends on:
  - ✓ expected changes in abnormal earnings;
  - ✓ next year's level of comprehensive earnings;
  - ✓ the cost of equity capital.

DETERMINANTS OF LEADING PRICE-TO-EARNINGS RATIOS

- The higher the expected changes in future abnormal earnings, the higher the PE ratio;
- The higher next year's comprehensive earnings, the lower the PE ratio (assuming non-negative expected changes in abnormal earnings);
- The higher the cost of equity capital, the lower the PE ratio (assuming non-negative expected changes in abnormal earnings).

DETERMINANTS OF LEADING PRICE-TO-EARNINGS RATIOS

- Consider a firm without any source of comparative advantage within a very competitive industry;
- Further assume that accounting is unbiased;
- Such a firm should trade on a leading (levered) price-to-earnings ratio equal to $1/r_E$ in efficient capital markets;
- This ratio is referred to as the normal levered leading price-to-earnings ratio.

APPLICATION EXERCISE 3

- Consider Caelum Inc. again;
- Calculate Caelum Inc's intrinsic (levered) leading PE ratio at the beginning of year 1.

APPLICATION EXERCISE 3

Year		**1**	**2**	**3**	**4**
AEi			29	35	0
ΔAEi			13	6	-35
PV(ΔAEi)			11.8	5.0	-26.3
PV(Sum of ΔAEi)/CE1	-0.38				
1/rE	10				
Leading PE	**6.19**				

JOINT DISTRIBUTION OF PB AND PE RATIOS STRATEGIC TAXOMONY

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JOINT DISTRIBUTION OF PB AND PE RATIOS

- Joint distribution of PB and trailing PE ratios:

		PB ratio	
		High	**Low**
PE ratio	**High**	32.8%	16.7%
	Low	17.1%	33.4%

STRATEGIC TAXONOMY

- Dogs;
- Falling Stars;
- Recovering Dogs;
- Rising Stars.

STRATEGIC TAXONOMY

- Rising stars:

  Firms expected to enjoy:

  - ✓ abnormally high AE in the future on assets in place;
  - ✓ abnormally high AE on future investments.

- Falling stars:

  Firms expected to enjoy:

  - ✓ abnormally high AE in the future on assets in place;
  - ✓ no significant AE on future investments.

STRATEGIC TAXONOMY

- Dogs:

  Firms expected to enjoy:

  - ✓ no significant AE in the future on assets in place;
  - ✓ no significant AE on future investments.

- Recovering dogs:

  Firms expected to:

  - ✓ rebound in the short-term from temporarily low earnings;
  - ✓ experience subsequently dog performance.

STRATEGIC TAXONOMY

	Low PB	High PB
Low PE		
High PE		